ALGEBRAIC SPACES SATISFY THE SHEAF PROPERTY FOR THE FPQC TOPOLOGY (GABBER)

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 022H: sources/stacks-chow/tags.txt, source lines 2706–2706. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02X4: sources/stacks-chow/tags.txt, source lines 3827–3827. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02L4: sources/stacks-chow/tags.txt, source lines 3395–3395. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03BX: sources/stacks-chow/tags.txt, source lines 4364–4364. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03FX: sources/stacks-chow/tags.txt, source lines 4508–4508. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03FZ: sources/stacks-chow/tags.txt, source lines 4510–4510. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03H4: sources/stacks-chow/tags.txt, source lines 4553–4553. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 023V: sources/stacks-chow/tags.txt, source lines 2756–2756. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original reference 0AP9: sources/stacks-chow/tags.txt, source lines 13467–13467. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03WS: sources/stacks-chow/tags.txt, source lines 5123–5123. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0APK: sources/stacks-chow/tags.txt, source lines 13478–13478. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02KV: sources/stacks-chow/tags.txt, source lines 3386–3386. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02VN: sources/stacks-chow/tags.txt, source lines 3774–3774. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 023Q: sources/stacks-chow/tags.txt, source lines 2751–2751. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1 (Gabber)**.** *Let S be a scheme. Let X be an algebraic space over S. Then X satisfies the sheaf property for the fpqc topology.*

Proof. Since X is a sheaf for the Zariski topology it suffices to show the following. Given a surjective flat morphism of affines $f : T' \to T$ we have: $X(T)$ is the equalizer of the two maps $X(T') \to X(T' \times_T T')$. See Topologies, Lemma 022H (there is a little argument omitted here because the lemma cited is formulated for functors defined on the category of all schemes).

Let $a, b : T \to X$ be two morphisms such that $a \circ f = b \circ f$. We have to show $a = b$. Consider the fibre product

$$E = X \times_{\Delta_{X/S}, X \times_S X, (a,b)} T.$$

By Spaces, Lemma 02X4 the morphism $\Delta_{X/S}$ is a representable monomorphism. Hence $E \to T$ is a monomorphism of schemes. Our assumption that $a \circ f = b \circ f$ implies that $T' \to T$ factors (uniquely) through E. Consider the commutative diagram

$$\begin{array}{ccc} T' \times_T E & \longrightarrow & E \\ \downarrow & \nearrow & \downarrow \\ T' & \longrightarrow & T \end{array}$$

Since the projection $T' \times_T E \to T'$ is a monomorphism with a section we conclude it is an isomorphism. Hence we conclude that $E \to T$ is an isomorphism by Descent, Lemma 02L4. This means $a = b$ as desired.

Next, let $c : T' \to X$ be a morphism such that the two compositions $T' \times_T T' \to T' \to X$ are the same. We have to find a morphism $a : T \to X$ whose composition with $T' \to T$ is c. Choose an affine scheme U and an étale morphism $U \to X$ such that the image of $|U| \to |X|$ contains the image of $|c| : |T'| \to |X|$. This is possible by Lemmas 03BX and 03FX, the fact that a finite disjoint union of affines is affine, and the fact that $|T'|$ is quasi-compact (small argument omitted). Since $U \to X$ is separated (Lemma 03FZ), we see that

$$V = U \times_{X,c} T' \longrightarrow T'$$

is a surjective, étale, separated morphism of schemes (to see that it is surjective use Lemma 03H4 and our choice of $U \to X$). The fact that $c \circ \mathrm{pr}_0 = c \circ \mathrm{pr}_1$ means that we obtain a descent datum on $V/T'/T$ (Descent, Definition 023V) because

$$\begin{aligned} V \times_{T'} (T' \times_T T') &= U \times_{X, c \circ \mathrm{pr}_0} (T' \times_T T') \\ &= (T' \times_T T') \times_{c \circ \mathrm{pr}_1, X} U \\ &= (T' \times_T T') \times_{T'} V \end{aligned}$$

The morphism $V \to T'$ is ind-quasi-affine by More on Morphisms, Lemma 0AP9 (because étale morphisms are locally quasi-finite, see Morphisms, Lemma 03WS). By More on Groupoids, Lemma 0APK the descent datum is effective. Say $W \to T$ is a morphism such that there is an isomorphism $\alpha : T' \times_T W \to V$ compatible with the given descent datum on V and the canonical descent datum on $T' \times_T W$. Then $W \to T$ is surjective and étale (Descent, Lemmas 02KV and 02VN). Consider the composition

$$b' : T' \times_T W \longrightarrow V = U \times_{X,c} T' \longrightarrow U$$

The two compositions $b' \circ (\mathrm{pr}_0, 1), b' \circ (\mathrm{pr}_1, 1) : (T' \times_T T') \times_T W \to T' \times_T W \to U$ agree by our choice of α and the corresponding property of c (computation omitted). Hence b' descends to a morphism $b : W \to U$ by Descent, Lemma 023Q. The diagram

$$\begin{array}{ccccc} T' \times_T W & \longrightarrow & W & \xrightarrow[b]{} & U \\ \downarrow & & & & \downarrow \\ T' & & \xrightarrow{\quad c \quad} & & X \end{array}$$

is commutative. What this means is that we have proved the existence of a étale locally on T, i.e., we have an $a' : W \to X$. However, since we have proved uniqueness in the first paragraph, we find that this étale local solution satisfies the glueing condition, i.e., we have $\mathrm{pr}_0^* a' = \mathrm{pr}_1^* a'$ as elements of $X(W \times_T W)$. Since X is an étale sheaf we find a unique $a \in X(T)$ restricting to a' on W. □